

Sweepback Effects on the Lift, Drag, and Pitching Moment Coefficients for a B-1 Bomber Model in a Subsonic Speed Regime

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The introduction of the variable swept wing design provided the capability of changing an aircraft's planform in flight. Unswept wings were generally used for lower speed flights, while swept wings provided flight capabilities in higher speed regimes; hence, changing the sweep settings allowed the aircraft to optimize for different mission requirements. Unfortunately, accurate predictions for the aerodynamic characteristics for an aircraft with a variable swept wing design, such as the B-1 Bomber, is difficult. As a result, acquiring data from wind tunnel tests is critical in order to accurately determine the optimal angle of attack and corresponding sweep angle settings in order to maximize C_L and minimize C_D on the B-1 Bomber; this becomes the overall objective of this experiment.

Nomenclature³

AOA	=	angle of attack
C_L	=	aircraft lift coefficient
C_D	=	aircraft drag coefficient
C_M	=	aircraft pitching moment coefficient
WSU	=	Wichita State University

I. Results Summary¹

Experimental analysis of a 1:48 scaled B-1 Bomber model is presented in this report. The model was tested at 5 different sweep angles as measured to the leading edge of the wing: 21°, 32°, 43°, 54° and 65°. In addition, alpha-sweeps from -8° to 16° were considered, where data was acquired in 4° increments. Aerodynamics forces and moments were recorded using the data acquisition software in WSU's 3' X 4' wind tunnel.

The analytical results from testing demonstrated that sweeping the wing back will reduce the drag, though a compromise must be made with the lift generated. Pitching moments varied significantly as the aircraft center of pressure translated aft when the sweep is increased; this limits the alpha-sweep conducted, since the pitching moment limits of the wind tunnel is a constraint that must be taken into account. The performance of the model (i.e. the lift-to-drag ratio) did decrease with increasing sweep angles, as predicted.

The B-1 Bomber aircraft operates under much higher Reynolds Number in comparison to the Reynolds number for the B-1 model when operated in the wind tunnel. However, the results obtained are useful in terms of predicting the performance of a variable swept wing aircraft within the subsonic regime. The model was tested at 0.14 Mach, hence, the flow can be assumed to be steady, incompressible and irrotational.

II. Repair and Construction²

A. Model Specifications

Utilizing a B-1 Bomber model for testing purposes was decided on since it incorporated a variable swept wing design, and because it was readily available at WSU. Prior to testing in the wind tunnel, dimensions from the model

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were recorded for calculation purposes, and to ensure that the model was indeed scaled 1:48 from the actual aircraft. Results from these measurements are presented alongside the actual aircraft specifications in **Table 1**.

Table 1: Specifications for the B-1 Aircraft and B-1 Model

Characteristics	Real Aircraft (Ref. 1B-1 Handbook)		Model	
Length	147 ft		36.75 in	
Wingspan	Extended	136.7 ft	Extended	34.175 in
	Swept	78.2 ft	Swept	19.55 in
Sweep Angle	Fully Extended	15°	Fully Extended	21°
	Fully Swept	65°	Fully Swept	65°
Height	34 ft		2 in.*	
Weight	Empty	192,133 lb	12 lb	
	Max. Takeoff	477,000 lb		
Max. Speed	825 mph		110 mph	

*Height was considered without landing gear height since this component was not included with the model.

B. Model Repair

After confirming the model was 1:48 scale, repairing the model became a priority for safety reasons, and in order to obtain accurate wind tunnel results. Initial signs of disrepair included chipped edges on the empennage and wing surfaces, a missing nozzle on an engine which had become detached from the model aircraft, open holes, and dented surfaces on the model. Bondo was used to fill the open holes, to build dented areas, to reconstruct the missing nozzle, and to fix the chipped edges on the empennage and wing. The bondo was allowed to cure for at least 24 hours before it was sanded down into a smooth surface.

Once this was completed, the model was painted with two coats of paint. The first coat acts as a protective layer on the model, while the second coating ensures a smooth surface.

C. Securing Sweep Angles

In order to secure the wing sweep angle during testing, two solutions were considered. The first solution consisted of securing the wing with aluminum tape at the wing root chord. Unfortunately, after testing this solution, the aluminum tape was found unable to sustain the shear stresses when minimal force was applied at the wing tip.

The second solution consisted of securing the wing by pinching the wing, whereby a screw is secured near the pivot point locations on the wing. When a force is applied at the wing tip, the screw also failed to secure the wing. Since the flat surface near the pivot point was resting against the flat end of the screw, an insufficient friction force was present, which proved unable to stop the wing from rotating. As a result, electrical tape was attached to the end of the screw, and adhered to the flat surface of the pivot point, thus securing the sweep angle (see Fig. 1). When a force is applied on the wing tip, the wing was rigid, and maintained a constant sweep angle.

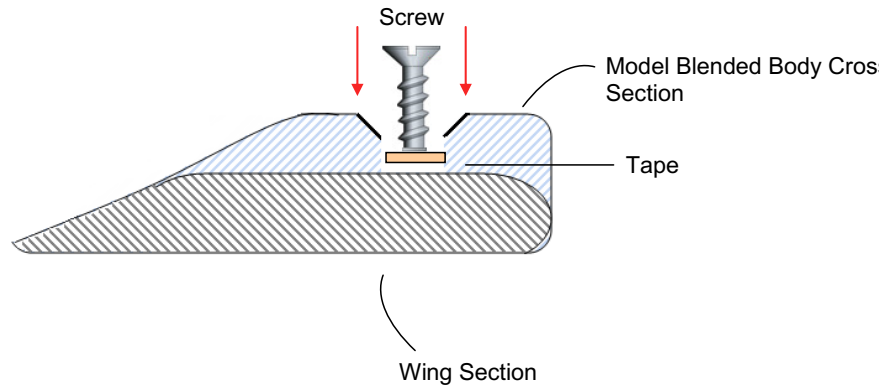


Fig. 1: Securing Mechanism – Cross-Section View. When the desired sweep angle is positioned, a screw with electrical tape attached to the end will be fastened to the model. The tape creates additional friction which arrests the wing's movement.

Concurrently during this time, the desired sweep angles were marked on the wing's surface (see Image 1). A difference was noticed between the wing fully extended sweep angle on the B-1 aircraft, and the model's sweep angle with fully extended wings. The aircraft has a fully extended wing with 15° of sweep built into the design, whereas the model contains 21° of sweep. Hence, testing the model's sweep angles was limited to an initial value of 21° with 11° increments to the fully swept wing angle of 65° .

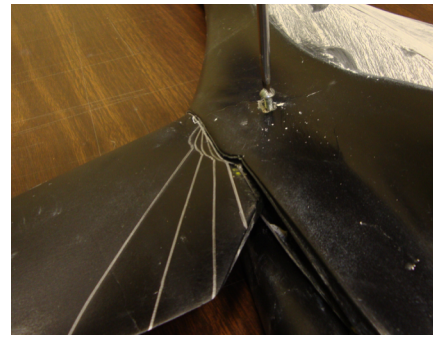


Image 1: Desired swept angle and securing location.

D. Construction of the Mounting Plate

The mounting plate was constructed to meet the dimensions of the mounting plane of the wind tunnel balance system (see Fig. 2). The mounting plate was secured at two existing screw holes located on the bottom surface of the model as shown in images 2 and 3. The mount was constructed such that the balance reference center would remain as close as possible to CG of the aircraft. Image 3 shows how the model was mounted to the balance system with only 4 screws; the middle 2 screws and the 2 aft-most screws. Upon mounting the model in the tunnel, it was discovered that the fore-most 2 screws would contact the balance at higher angles of attack, causing part of the load to be transferred to balance system.

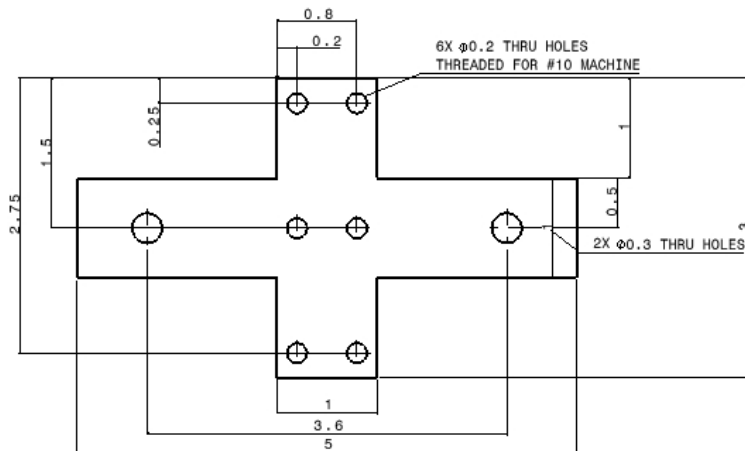


Fig. 2: CATIA drawing of mounting plate specifications. All measurements shown in inches.



Image 2: Marked location for securing mounting plate.

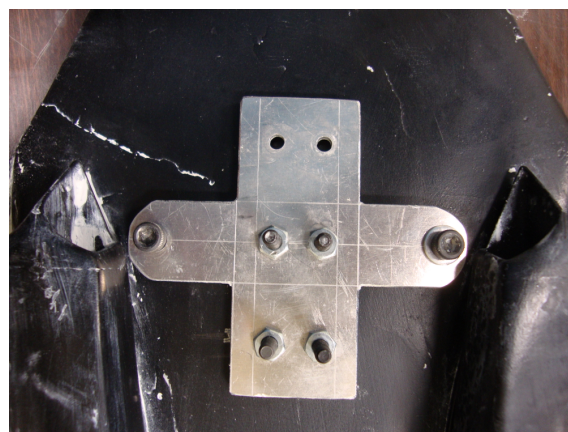


Image 3: Secured mounting plate on the model.

III. Test Matrix⁴

Our test runs were made at the WSU 3x4 Open Wind Tunnel at a range of Q's between 20 and 29 with a special focus on the highest Q possible and changing the AOA between -8 and +16°. A total of 35 runs including dynamic tares, static tare, flow visualization, and runs with and without flow trips were completed.

After this, we decided to go ahead and adjust the model to the support system with only four of the six bolts since the ones on the front could get stuck with the mounting system when changing AOA. The wing's sweep angle was hold with round end bolts on the upper part of the fuselage.

We started our normal runs with the fully extended configuration or 21° starting with Q=20 and going from -8 to +16 AOA, after this set was done we increased the Q to 25 and 29 with the corresponding evaluation test at each one. After these runs were done we made the same process for the following sweep angles: 34°, 43°, 54°, and fully swept or 65°. A total of 15 normal runs were made.

Once we were done with the normal runs we added to our model pieces of metal tape on the nose, on the upper and lower part of the wing's leading edge, and also on both sides of the tail in order to get a more continuous flow along the model's surface. Runs made with this process are called flow trips runs. These runs help us get better data since the flow over our model separates further than the normal runs. Tests with this configuration were made at the highest Q possible since our focus was on high speed flow. A total of 5 flow trips runs were made, one per each sweep angle with AOA between -8 and +16.

The last runs of our test were flow visualization runs. For flow visualization we used china clay and kerosene since this mix allows the view of flow condition over the surface of the model. We made initial runs using flow trips and china clay-kerosene mix, but since most of our data was obtained from the normal runs we decided to finish out tests with no flow trips and china clay-kerosene mix. These tests were limited due to time and material and these were the reasons why we tested just two sweep angles with two different AOA at the highest Q possible. It is important to remember that china clay-kerosene mix dries very fast and that the flow visualization results can be damaged with a simple touch since once dry it is similar to chalk. Also since china clay-kerosene mix is only good for one configuration with a single AOA, each different AOA or configuration was count as one run. We made 7 runs of flow visualization: two for 65° sweep angle, flow trips and AOA of 0 and +16; two more for 21° sweep angle, no flow trips and AOA of 0 and +16, and three runs of fully swept of 65° sweep angle, no flow trips and AOA of 0, -8 and +16.

Our observations for the flow visualization were that we had one big vortex near the leading edge of the wing going almost parallel to its shape. On the tail wing we observed that on its leading edge there was a flow separation since no flow was present, but about a 1/3 later flow was very smooth. On the lower part of the model after the cockpit, flow was going up on the sides creating on the upper surface a single smooth flow.

IV. Predicted Results and Test Results³

E. Predicted Results

Initial predictions for the lift was undertaken to ensure that the wind tunnel lift load limit of 35 lbs was not exceeded. In order to predict the lift values of the model, the lift was calculated utilizing airfoil data from *Theory of Wing Sections*. Two airfoils were chosen; one for the wing-body, and one for the horizontal tail. Since the horizontal tail of the model closely resembled a flat plate, the symmetrical airfoil NACA 0006 was utilized to predict the lift contribution from the horizontal tail.

When calculating lift for the wing-body, the area of the fuselage is included in the wing area. In this way, the lift contribution from the fuselage can also

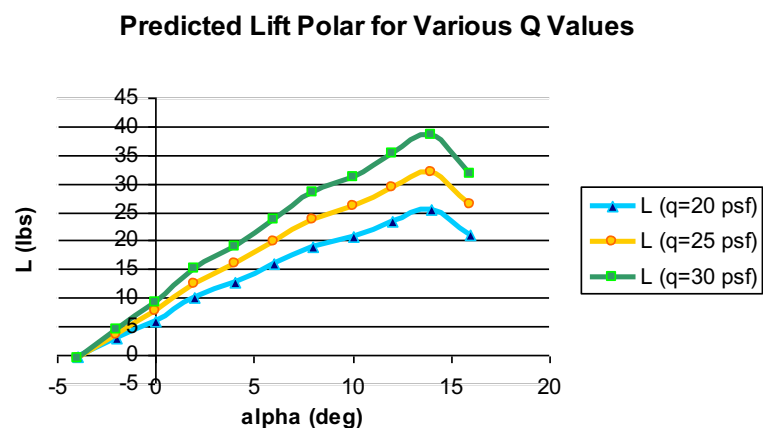


Fig. 3: Predicted lift behavior as a function of Angle of Attack.

be taken into account. Data from the NACA 4415 airfoil was utilized to predict the lift contribution from the wing-body. Prediction results are shown in Fig. 3.

Observe how the lift generated at a dynamic pressure of 30 psf does exceed 35 lbs of lift. This was not seen as a major source of concern since the airfoil data was extrapolated at a Reynolds number of 300,000, whereas, the Reynolds number for the model is closer to 80,000. Hence, unless the lift generated was much larger than 35 lbs, such as lift values exceeding 50 lbs, predictions indicate the lift generated will not exceed wind tunnels lift load limit. Results from testing confirmed the lift load limit was not exceeded (see **F. Test Results**).

F. Variation of Predicted Results with Sweep Angle

Since the B-1 has a complex fuselage geometry, and additional components such as the engines and sensors, accurate predictions for the variation of lift, drag, and pitching moment with different sweep angles would have been difficult to model. Instead, the relative trend of lift and drag behaviors as the sweep angle is increased was researched.

Anderson states that “for subsonic flight, increasing the wing sweep reduces the lift” (pp. 331). Hence, the curves in Fig. 3 are expected to translate further such that lift values will be closer to the abscissa as the sweep angle varies. This was validated during testing, and can be seen in the lift polar provided in Fig. 6.

Anderson also presents the variation of the lift-to-drag ratio with sweep angle shown in Fig. 4. One can observe how the lift-to-drag ratio decreases dramatically as the sweep angle is increased from 40° to 60° sweep. Since the B-1 model has a maximum sweep of 65°, only values from 20° to 65° were extrapolated from Anderson, and compared to the test results.

General behaviors for drag and pitching moment could not be found during the research; hence testing results were analyzed without prior predictions.

Test Results

When a comparison is made between the results that Anderson presents and the collected data from the experiment, one can see how the trend of the curves are fairly close, signifying the concurrence of the test results with historical results (see Fig. 5). In addition, one will notice how the test result curves are translated a great distance from Anderson’s results. This is due to the speed differences under which Anderson’s tests were conducted as compared to the speed under which this experiment was operating under. From Fig. 4, comments within the figure indicate the test was conducted at 0.6 Mach. In comparison, for a maximum q value of 29 psf, the maximum speed found in this experiment was 0.14 Mach, a significant difference.

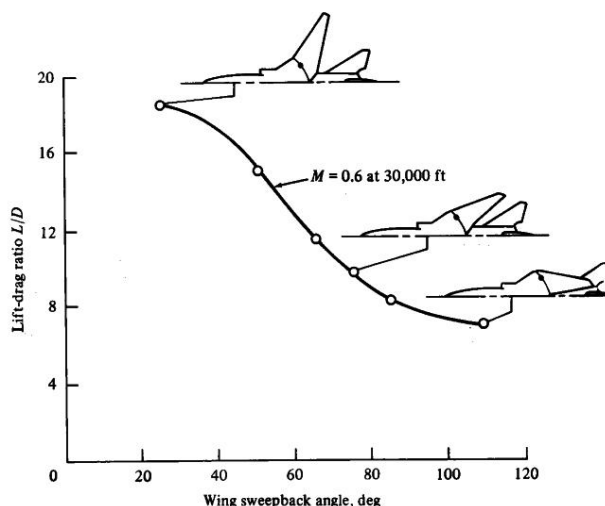


Fig. 4: Variation of lift-to-drag ratio with sweep angle as presented by Anderson (Ref. 3).

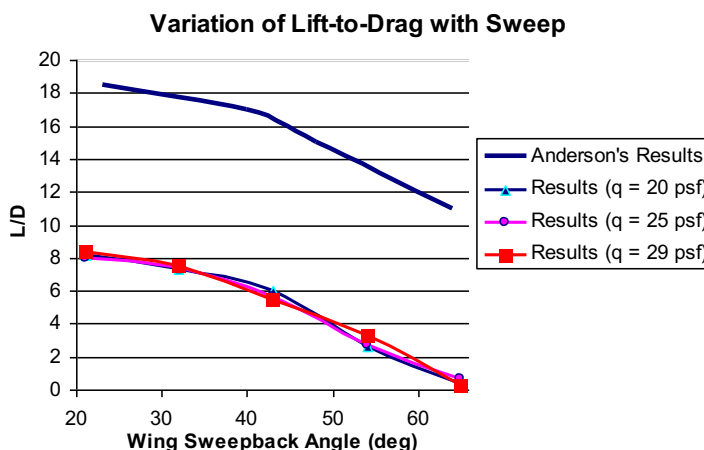


Fig. 5: Comparison of test results with Anderson’s results for the variation of the lift-to-drag ratio with sweep angle.

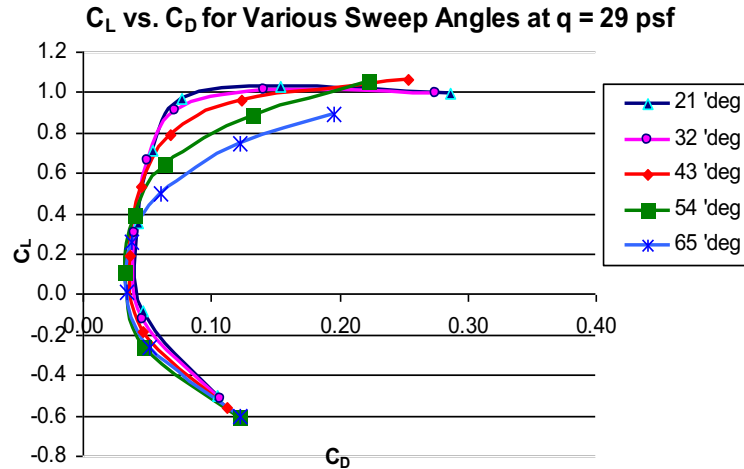


Fig. 6: Comparison of C_L values for each C_D value found for each sweep angle tested.

When the results are reduced into C_L and C_D data, and plotted, the observation can be made that greater sweep angles result in lower values of C_L for each C_D considered. Recall the objective of this experiment was to determine the optimal angle of attack and corresponding sweep angle settings in order to maximize C_L and minimize C_D on the B-1 Bomber within a subsonic speed regime. From Figure 6, the results indicate the optimal setting in order to achieve maximum C_L values for minimal C_D values would include utilizing a wing which incorporates 21° of sweep as measured to the wing leading edge. In order to determine what the optimal angle of attack would be, a plot of the lift polar

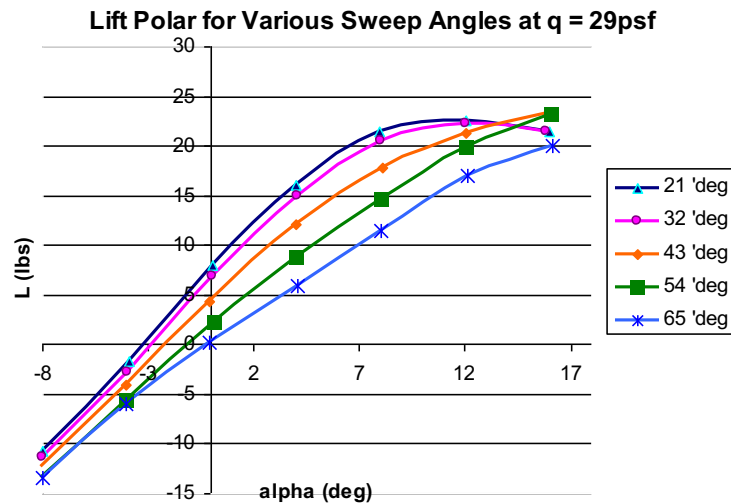


Fig. 7: Lift polar for each sweep angle considered during testing.

for the various sweep angles was plotted (see Fig. 7). From Figure 7, one can observe stall is beginning at an AOA of 8° when 21° of sweep is considered. Hence, the optimal angle of attack would be 8° .

After considering the lift and drag behavior of the model in test results, the pitching moment was analyzed. Figure 8 demonstrates how increasing the sweep angle causes the pitching moment to increase. This is logical, since sweeping the wing back shifts the aircraft center of pressure further aft. An interesting phenomenon can be seen when comparing Figure 7 & 8 concurrently. Notice how at higher sweep angles (e.g. 54° and 65° sweep) the slope of the pitching moment polar

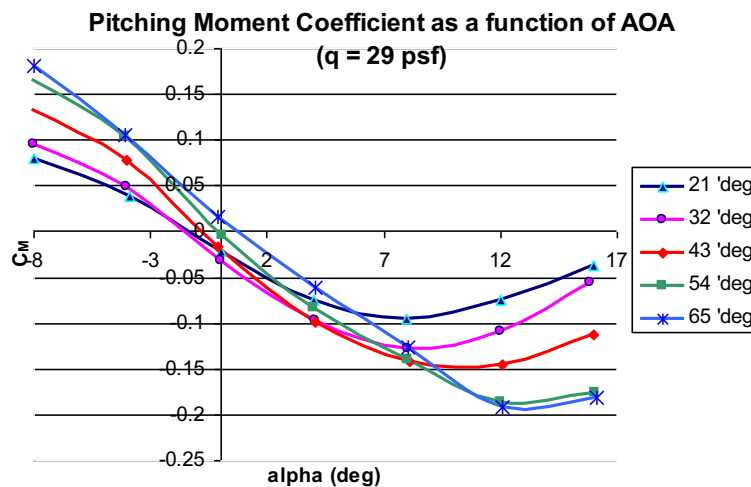


Fig. 8: Pitching moment coefficient is considered about the calculated aerodynamic center for the model.

becomes positive; indicating the approach of stall, however the lift curves for these higher angles of sweep remains fairly linear during this time. Flow visualization images explain these increases in lift as being the result of vortex induced lift (see IV. **Flow Visualization** for more details.)

G. Correction Factors and Uncertainty

In order to correctly interpret the results from wind tunnel tests, several correction factors must be made. These corrections include removing static and dynamic tare effects, translating the pitching moment from the balance system's

reference center to the calculated aerodynamic center, and accounting for the model's solid blocking effects.

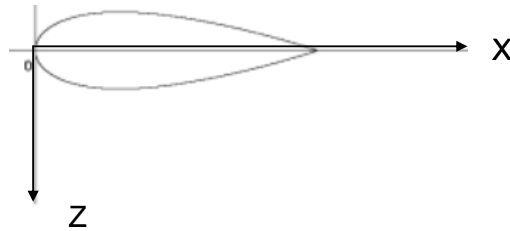


Fig. 9: Reference axis considered at the leading edge of the wing root chord for aerodynamic center calculations.

Calculation

of the model aerodynamic center was undergone with the following assumptions: the distance between the model reference center and the actual aerodynamic center location along the z-axis is small, hence any corrections along the z-axis is neglected (see Fig. 9). In addition, since the AOAs tested are small values, $\sin\alpha$ is assumed to equal zero, and $\cos\alpha$ is assumed to equal one.

Solid blocking corrections were made using Thom's solid blocking correction equation (pp. 320 LSWT).

In addition to these corrections, standard deviations for lift, drag and pitching moment were given through the data acquisition software. These standard deviations were reduced in terms of percent uncertainty for each test run, and are summarized in Table 2 below.

Table 2: Standard Deviations Expressed as Percent Uncertainty averaged over Each Set of Runs operating at similar q values.

q (psf)	L (% Uncertainty)	D (% Uncertainty)	PM (% Uncertainty)
20	3.89%	27.15%	17.12%
25	5.12%	28.51%	18.63%
29	5.87%	28.05%	17.95%

V. Additional Test Runs

H. Flow Trips¹

Flow tripping is a method utilized to simulate flight conditions during slow speed wind-tunnel testing. Aluminum tape was serrated and then attached to the leading edges of the wing, tails, and the nose in order to trip the flow. The model was then tested at the maximum 'q' possible. Flow visualizations were done using the flow trips in order to observe the boundary layers. Image 10 shows the effects of tripping the flow, on the boundary layer. As a result of flow tripping we can see that the C_{Lmax} has increased compared to the flow without trips, but there is also a significant increase in the C_D (see Fig. 10 & 11). At a sweep of 54° the model didn't stall during the flow trip run even when we pitched it about 16° angle of attack. Flow visualizations showed that there were vortices being generated on the leading edge of the wings resulting in vortex-lift being generated, which explains the delay in the stall.



Image 4: Attached turbulent flow for tripped flow.
Configuration: Angle of attack of 16° ,
Dynamic Pressure of 29 psf

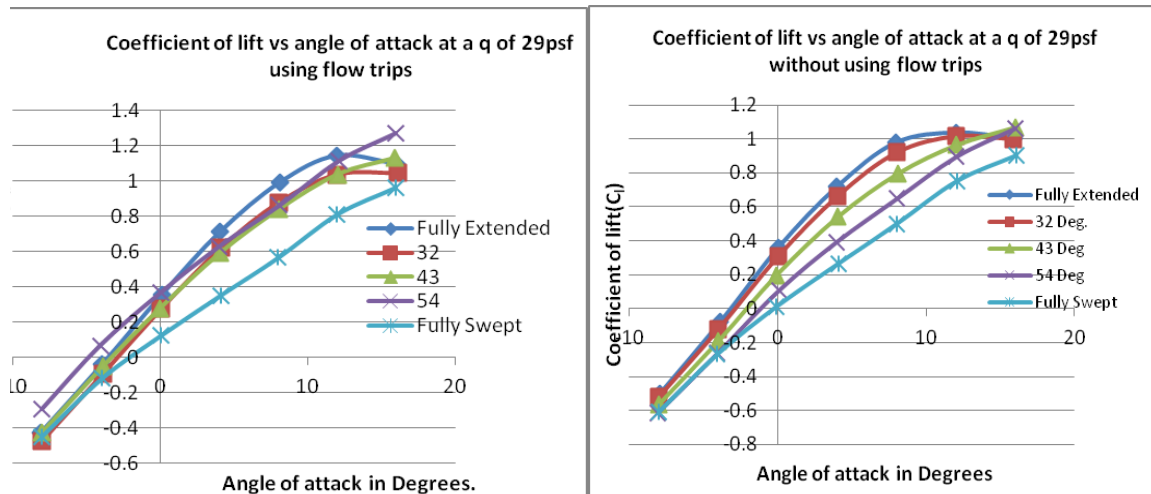


Fig. 10 & 11: Comparison between Lift Polar Curves for Runs with and without Flow Trips. The attached turbulent layer for runs with flow trips causes a greater lift production. Hence, in actual flight conditions the lift generated would be greater than the values obtained in the wind tunnel for runs without flow trips.

I. Flow Visualization²

China clay and kerosene were used for the visualization test. Since the china clay and kerosene compound dries quickly, only half of the model is coated. Coating the model required brushing the compounds on in a crosswise motion perpendicular to the air flow, thus allowing one to differentiate between brush strokes that have dried in the coating with the air flow effects over the model.

Due to time constraints, select visualization tests were conducted, all of them at maximum dynamic pressure of 29 psf. See Table 3 & 4 for flow visualization results.

Table 3: Flow Visualization for Wing Fully Extended (21°)

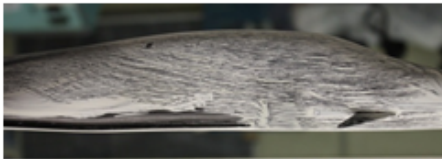
Fully Extended (21°)		
Results	AoA	Description
	0°	Nice and steady air flow along the entire model body at this AoA compared to 16° AoA.
	0°	The Transition between laminar and turbulent boundary layer can be clearly seen on the picture. The separation on the flow is due to lack of energy. Flow tripping would attach the turbulent boundary layer to the surface for much longer.
	16°	Flow separation and reverse flow can be seen on the model surface at higher angle of attacks, which results in huge drag increment and stall on the model.

Table 4: Flow Visualization for Wing Fully Swept (21°)

Fully Swept (65°)		
Results	AoA	Description
	16°	Formation of vortices began from the bottom of the nose at high AoA.
	0°	Larger laminar flow on the wing at fully swept angle (65°) compared to the fully extended angle (21°). This also caused reduction in significant amount of drag.
	16°	Leading edge vortices have formed, compare this to the flow separation observed when the wing is fully extended at high AoA. The horizontal tail shows flow reversal effects. At this configuration the model hasn't stalled though the tail shows signs of stall because of the induced vortex lift.

VI. Concluding Remarks¹

J. Conclusion

A 1:48 scaled B-1 model was successfully tested in the WSU 3'x4' wind tunnel. The data was then reduced to obtain different aerodynamic properties like lift, drag, pitching moment etc. The data was further reduced to dimensionless coefficients such as Lift Coefficient (C_L), Drag Coefficient (C_D) and Pitching Moment Coefficient (C_M).

Sweeping the angle backwards significantly reduced the drag, hence this kind of configuration is highly recommended for higher speeds and cruises, whereas a fully extended wing produces more lift hence this is recommended for takeoffs.

The pitching moment increases rapidly as the wings are swept backwards because of the fact that the center of pressure of the aircraft actually moves backward from the aerodynamic center.

The Lift/Drag ratio decreases as we sweep the angle backwards but at higher speeds ($M=0.6+$) the drag divergence Mach Number increases with the increase in the sweep angle².

Flow tripping simulated flight conditions and we saw significant increase in the lift and drag.

Flow visualization gave a better picture of flow behaviors like the transition of boundary layer, vortex generation, flow separation etc.

K. Suggestions For Further Work

The same experiment can be done at higher speed to get the effects of variable sweep on drag divergence and other important aerodynamic aspects.

A real model could be built with proper airfoils and body and some proper predictions could be made on flow behavior and the results itself.

The subsonic data can be used to extrapolate supersonic results considering compressible flow and performing numerical analysis.

Wing-Body interference can be calculated and hence more precise results can be obtained.